Mohit Appari

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EXPERIENCE

Agency of Persons with Disabilities - State of Florida, *Data Scientist, Tallahassee, FL* Jun 2025 - Present

- Analyzed millions of transactional records representing ~$200M in average monthly spending across 60K+ clients, integrating data from SQL Server and internal systems to deliver financial analysis supporting budgeting, expenditure planning, and operational decision-making.
- Developed spending and budget forecasts using ARIMA, regression, moving averages, and time-series decomposition, modeling historical utilization and spending trends to produce projections typically within 5–10% of actual expenditures.
- Proposed reimbursement-rate changes across 22+ service areas, analyzing historical spending and utilization to quantify projected costs under alternative rates and delivering data-driven recommendations to leadership on budget feasibility, additional funding needs, and provider impact
- Performed weekly claims-to-payment validation and reconciliation, analyzing billed versus paid amounts to monitor financial accuracy, identify discrepancies, and support reliable reporting across high-volume transactional data.
- Re-engineered recurring SPSS analyses and reporting workflows in Python, cutting analysis turnaround time from ~2 hours to under 30 minutes (75% reduction); optimized SQL queries and pipelines with indexing, partitioning, and caching, slashing report execution time by 83% (60 minutes to under 10 minutes)

Florida State University, *AI/ ML Researcher, Tallahassee, Florida* June 2024 - Sept 2024

- Fine-tuned and benchmarked NequIP, an equivariant neural network, on a custom dataset of over 11 000 material structures, optimizing hyperparameters and training settings, which improved material property prediction accuracy
- Developed and evaluated custom weighted loss functions using MAE and MSE to balance multiple prediction objectives, reducing validation error by ~15%.
- Built reproducible Python workflows using Docker and MLflow for data preparation, model training, experimentation, and evaluation, and collaborated with PhD researchers to turn research requirements into quantitative ML models, resulting in faster experiment cycles and consistent results

S&P Global, *Software Developer, Bangalore, India* Jan 2022 - Aug 2023

- Identified and led the migration of document-processing workflows from licensed .NET tooling to a Python-based extraction platform, evaluating open-source OCR and document-processing technologies and reducing licensing costs by 80%.
- Led a 5-person engineering team through the architecture, development, and production rollout of the Python extraction platform, processing 10,000+ financial, energy, and automotive documents from a single project area in under 5 minutes, while achieving 95% extraction accuracy across the broader workflow.
- Engineered fault-tolerant AWS/Docker pipelines with event-driven ingestion, automated retries, structured logging, and validation, enabling unattended processing and reducing manual extraction effort by 90%+.
- Applied OCR and CNN-based classification to extract structured data from complex unstructured documents, including financial filings and earnings information, energy specifications, and automotive parts attributes, improving downstream entity-recognition accuracy.
- Designed modular, reusable components and automated CI/CD workflows with GitHub Actions, maintaining 80% test coverage and improving reliability and maintainability of the production platform.

LG Electronics, *Software Developer Intern, Bangalore, India* May 2021 - Aug 2021

- Built a real-time driver drowsiness detection system using Python, OpenCV, YOLO, and Raspberry Pi, achieving 90% facial landmark detection accuracy and demonstrating an end-to-end computer vision pipeline for safety-critical applications.
- Developed a CNN-powered logistics tracking system integrated with REST APIs and PostgreSQL, reducing package sorting time by 25% and earning 3rd place in LG's internal Ideathon competition.

ITC Limited, *Software Developer Intern, Bangalore, India* Jan 2021 - Mar 2021

- Developed a Python/Flask and MySQL master data management system with role-based access control and approval workflows, improving data security and operational efficiency across multiple business units.
- Automated product tracking and data-update workflows, reducing manual workload by 30% while improving data consistency and accuracy across business operations.

EDUCATION

Florida State University, *Masters in Data Science, Tallahassee, FL* Aug 2023 - May 2025

- *Cumulative GPA:* 4/4
- *Coursework:* Statistics, Mathematics, Research, LLMs, Analytics, AI, Databases, Big Data, Business and Finance, Akuna 101 Options Certification

Christ University, *Bachelors in Computer Science, Bangalore, IN* Aug 2018 - May 2022

- *Cumulative GPA:* 3.85/4
- *Coursework:* Data Structures, Operating System, Software Architecture, Programming, Networking, Communication, SDLC

PROJECTS

SeekingAlpha - A Quantitative Equity Research & Backtesting Platform | Python, DuckDB, FastAPI, Next.js, yfinance, Alpaca

- Built a quantitative equity research and backtesting platform covering 500+ securities and 2+ years of market data, combining fundamental, technical, momentum, and event factors with cross-sectional normalization, composite scoring, risk metrics, and automated portfolio analysis. Read Here.

BayWheels Lyft Trip Analytics | Python, Prefect, Apache (Spark, Kafka), Docker, Terraform, GCP (BigQuery, GCS), dbt

- Built an end-to-end batch and real-time analytics platform processing 400K+ monthly ride records using Kafka, Spark, BigQuery, and dbt, with Terraform, Prefect, and Docker for automated cloud infrastructure and workflows, reducing manual processing by 90% and reporting time by 60%.

STA5635 RAG App | Python, LangChain, OpenAI, Cohere, Pinecone, Flask, FastAPI, Streamlit, Docker, GitHub, Vercel

- Built a production-style RAG application for contextual Q&A over academic documents, integrating semantic retrieval and LLM-based generation through a vector database and API services to reduce research lookup time by 70%.

SKILLS

Programming: Python, R, SQL, Rust, C, C++, Java, JavaScript, TypeScript, HTML, CSS, Bash, Linux, PySpark, Go, Scala
Databases: PostgreSQL, MySQL, Oracle, MongoDB, Neo4j, Firebase, Redis, Cassandra, DuckDB, Snowflake, BigQuery, Delta Lake, Pinecone,
Cloud: AWS (EC2, S3, Lambda, RDS, SageMaker), GCP (BigQuery, Vertex AI), Azure Databricks, Kubernetes, Docker
Frameworks: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Prophet, Flask, FastAPI, Django, Streamlit, Dash, React, Node.js
Other Tools: Github, MLflow, Airflow, dbt, Kafka, Spark, Hadoop, Terraform, n8n, OpenAPI, Cohere, Ollama, Cursor, Langchain, Langgraph, Codex